TSMC

CoWoS and advanced back-end updates

Overweight

2330.TW, 2330 TT
Price (08 Apr 26):NT$1950.0
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Technology and Telecoms

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- **Raise CoWoS and SoIC capacity estimates:** We raise our CoWoS estimates slightly and now expect TSMC to have 115K/155K/175K wfpm CoWoS capacity by end-26/end-27/end-28. We also expect non-TSMC CoWoS capacity to slowly ramp up to ~25K/35K wfpm by end-27/end-28. CoPoS (or panel-level packaging) is likely to be the focus of capacity expansion in 2028 and beyond, but we do not yet build it into our forecasts, as the technology is still in the engineering and pilot production stage. We also raise our SoIC capacity expectations materially and now expect SoIC capacity to reach 35K/65K wfpm by end-27/28, to reflect growing demand from 2nm class AI-ASICs and incremental demand from Silicon Photonics/CPO adoption. We believe multiple AI accelerators (TPU v9, Trainium 4, future generations of Meta MTIAs and Open AI ASICs, as well as some versions of NVDA FPGAs) are likely to use SoIC packaging from 2028 onward. Co-package Optics under TSMC's COUPE platform will also be an incremental driver for SoIC demand, given that TSMC will be using SoIC for EIC-PIC integration.
- **Trim NVDA 2026 estimates to reflect higher Blackwell and lower Rubin, no change to 2027 estimates:** We trim our 2026 CoWoS estimates for NVDA by 4% (now 675K wafers per year) to reflect higher Blackwell shipments (5.8M units, up from 4.3M previously) and a slower ramp for Rubin (2.2M units, down from 2.8M previously), due primarily to HBM4 availability issues. Overall, NVDA GPU shipments are still likely to grow faster than prior estimates, in our view, reaching 8.3M units, up 34% YoY. We raise our CoWoS estimates for 2027 by 3% and now model all Rubin Ultra GPUs on a 2-die per interposer basis, given recent changes. We expect 4-die per package to be the mainstream solution, and the choice of 2-die or 4-die per package is unlikely to be material to the CoWoS allocation (which is based on a Chip-on-Wafer capacity basis). Moving to a 2-die-per-interposer solution may be more efficient for GPU chip output, given better yields (~10 dies per CoWoS wafer, similar to Rubin). Overall, we believe 2027 CoWoS forecasts are likely conservative and could move up in the next few quarters if NVDA is able to secure sufficient quantities of other bottleneck components (N3P wafers, HBM4), especially given that build plans for Vera CPU appear quite conservative, given extensive usage in Vera Rubin GPU racks and NVDA's intent to sell standalone Vera CPU racks. We believe this could be largely leveraging OSAT CoWoS-R or equivalent capacity (mainly ASE and Amkor).
- **Early reads on Feynman - Using A16 and 3D SoIC for high-end versions, base version to stay on N2P:** Our research indicates that NVDA Feynman (in 2028E) could have multiple SKUs, with the high-end versions using A16 process node at TSMC and leveraging 3D SoIC packaging technology for stacking multiple logic dies (likely GPU and SRAM stacks, based on our early read). We believe there will also be base versions of the Feynman family of GPUs using TSMC N2P process node, and 2.5D packaging in a 2-die-on-interposer configuration, with the product mix decision likely depending on the

See page 11 for analyst certification and important disclosures, including non-US analyst disclosures.

maturity of TSMC's A16 process and 3D SoIC stacking. We expect Feynman to adopt CoPoS packaging, depending on technology maturity and yields.

- **Strong upward revisions to Broadcom CoWoS estimates, primarily for TPU v7/v8:** We meaningfully revise up our 2026/27 CoWoS forecasts for Broadcom to 250K/400K, stemming largely from upside to TPU units. We now expect TPU shipments to reach 4.3M/6.9M units in 2026/27, with meaningful upside for Broadcom's Ironwood (TPUv7) and Sunfish (TPUv8ax), given increased external demand from customers. We retain our forecasts for MediaTek (18K/60K CoWoS wafers in 2026E/27E, implying 0.4M/1.5M TPU v8x (Zebrafish) units). MediaTek is still facing small changes to its Zebrafish design and could see some changes to this shipment schedule if the issues are not completely settled by 1H26. For TPUv9, our supply chain checks indicate that Google's design teams appear to be working on two competing designs with largely similar functional designs - Pumafish from Broadcom, based on 3D SoIC and TSMC CoWoS-L, and Humufish from MediaTek, using TSMC 3D SoIC, but with 2.5D packaging moving to Intel EMIB-T. At this point, we are unclear about which project gets higher volumes, but MediaTek's project looks more risky to us, given the adoption of multiple leading-edge technologies, including: (1) high-speed Serdes, with MediaTek aiming for a 300G class Serdes spec (336GHz likely, in our view) vs. Broadcom 400G Serdes; (2) Intel EMIB packaging execution due to no track record with AI accelerators with multiple embedded silicon bridges; and (3) 3D SoIC packaging design, with tape-outs of multiple compute and I/O dies within a short design window.
- **Trainium3 lifetime volumes stronger, slightly pushed out into 2027:** We trim our CoWoS forecasts for Trn3 (at AWS/Alchip) by 13%, but raise our 2027 estimates by 19%, to factor in higher Trn3 demand, but also a slight pushout of volume ramp in 2Q26. We now expect lifetime volumes to reach 4.4M units, slightly higher than prior estimates, due to strong compute demand from Anthropic and internal AWS workloads. For Trainium 3, we see almost all CoWoS wafers being processed by either AWS or Alchip directly, with no third fabless player in the mix. Given tight demand and TSMC's reluctance to expand CoWoS-R capacity, we see the possibility of some Trn3 volumes being outsourced to ASE in 2027 if the company can successfully ramp production of CoWoS process with high RDL layer count (>6-8 metal layers).
- **Lowering AMD CoWoS numbers slightly for 2026E, no change for 2027E:** As suggested in our prior update, we have seen some delays at TSMC for MI450 (including 2nm top-die re-tape-out and HBM4 characterization challenges). As a result, we mildly trim our CoWoS estimates for 2026 (77K vs. 90K previously), while leaving 2027E volumes unchanged. AMD could see upside in 2027 if MI450 starts ramping smoothly, given that its reliance on N2 process node for MI450 offers more capacity upside than other accelerators (all competing for limited N3P capacity). Venice CPU is also likely to consume meaningful CoWoS and 2.5D packaging capacity in 2H26 and 2027, with the bulk of the volume likely moving to ASE in 2027; this could be a source of upside to AMD CoWoS consumption in 2027, given increased datacenter CPU demand. AMD appears to also have multiple CPUs and gaming GPUs utilizing 2.5D packaging in its 2027 pipeline, with most of the wafer-based solutions using ASE and one panel-based design leveraging PTI.
- **Other ASICs largely unchanged, MTIA still slow, MAIA seems more optimistic into 2027:** We are not seeing a meaningful ramp for any of the other AI ASIC projects in 2026, with MTIA volume forecasts remaining quite tepid and supply chain preparation for 2027 also remaining lukewarm. MAIA 200 appears to have better-than-expected initial traction, and current forecasts indicate a strong ramp in 2027.
- **Increased CoWoS outsourcing for OSATs, EMIB likely to ramp only in 2028:** Given that TSMC's CoWoS capacity is already quite full and indicative customer demand is still 15-20% higher than supply, we continue to see a high level of engagement with OSATs on several CPU/accelerator projects. We expect ASE to start supporting AMD Venice and

NVDA Vera CPU projects in 2H26, with potential entry into Trn3 and TPU in 2027, if qualification is successful. For Amkor, we believe early engagements will start with NVDA GB10, Windows ARM CPU projects, Vera CPU and Broadcom Tomahawk 6 Switch IC. For Intel EMIB, we believe MediaTek's TPUv9 (Humufish) project is likely the first AI ASIC ramp, with volume production more likely in 2028 if technical challenges related to large package size and numerous silicon bridge integrations are resolved. Engagements with other ASIC customers remain quite minimal, based on our supply chain research.

- **CoPoS engineering work in progress, likely to be adopted in 2028 and to eventually replace CoWoS:** Our research indicates that TSMC and ASE are progressing on their engineering work on CoPoS, with pilot line likely ready by 1H26. We believe there are still engineering challenges that need to be overcome, with the aim of finalizing the process flow and equipment list by end-26. We believe CoPoS is slated to be adopted in 2028 (likely some versions of NVDA Feynman at the earliest) if engineering and pilot production proceed without any delay, and should eventually replace CoWoS for large-size AI accelerator packages. At this point, we still believe there are multiple engineering challenges that need to be overcome (warpage, equipment compatibility, material selection, etc.) before the process can enter mass production. CoWoS will still be used for smaller package sizes, with other processors (server CPU, networking, even PC and mobile processors) starting to adopt CoWoS-like packages in the 2028-29 timeframe.
- **CoWoP progressing on a parallel track, timing of introduction uncertain:** CoWoP, an NVDA-led effort to simplify the CoWoS packaging structure, is progressing on a parallel track. Our recent supply chain checks indicate encouraging progress on engineering milestones so far between NVDA, OSAT partners such as ASE and PCB vendors (Unimicron, ZDT, etc.). Key benefits of CoWoP include: (1) a more elegant package design that can eliminate the substrate layer completely; (2) hence increasing the signal integrity; (3) better thermal performance; and (4) much lower testing time, where increased chip probing and board-level testing coverage can eliminate significant amounts of final test and SLT. We are still not certain of the timing of introduction into mass production; we believe a small volume ramp of Rubin/Rubin Ultra 2-die solutions will be trialed using CoWoP at end-26/1H27 before any call on mass production ramp is made.

Note: Alphabet (GOOGL, OW), Amazon.com (AMZN, OW) and Meta Platforms (META, OW) are covered by Doug Anmuth. Advanced Micro Devices (AMD, N), Broadcom (AVGO, OW), Marvell Technology (MRVL, OW) and NVIDIA (NVDA, OW) are covered by Harlan Sur. Amkor Technology (AMKR, OW) is covered by Peter K. Peng. Apple (AAPL, OW) is covered by Samik Chatterjee, CFA. Microsoft (MSFT, OW) is covered by Mark R. Murphy. Tesla (TSLA, UW) is covered by Ryan Brinkman. Unimicron (3037 TT, OW) is covered by Jerry Tsai.

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Table 1: CoWoS allocation breakdown

CoWoS Consumption by customer (k wafer per year)	2023	2024	2025	2026E	2027E	JPM Comments
NVIDIA	70	192	400	675	1000	Downtick in Rubin due to HBM4 issues, uptick in GB300, overall 2026 numbers slightly lower due to better CoWoS efficiency. Vera CPU also to embrace CoWoS-R in 2026. 2027 growth mainly from Rubin and Rubin Ultra. LPU40 will use CoWoS-R from 2028.
At TSMC	*70*	*127*	*385*	*645*	*890*	2026 CoWoS -L at TSMC, others at OSAT, 2027 Rubin Ultra assuming 2-die-on-interposer solution.
CoWoS-S / R	*70*	*97*	*30*	*0*	*0*	Vera CPU using CoWoS-R. Start with TSMC but will eventually migrate to OSATs from late 2026.
CoWoS-L	*0*	*30*	*355*	*645*	*890*	Majority of CoWoS to migrate to CoWoS-L by 2H25. Rubin Ultra likely to remain on CoWoS-L, and could be the last CoWoS-based GPU before migrating to Panel-level for Feynman in 2028.
Outside TSMC (CoWoS-S/R)	*0*	*65*	*15*	*30*	*110*	Outsourcing ratio picking up in 2026 again as Vera outsourcing to ASE/SPIL and Amkor starts to pick up more momentum in 2027. Not seeing meaningful H200 production so far, in the supply chain.
AMD	10	45	52	77	120	MI300 and MI325 in 2024, MI355 in 2025; MI450 ramp in late 2026; moving to CoWoS-L for MI450 onwards; volume forecast includes Xilinx; not including China-specific SKUs in 2H25 and beyond.
CoWoS-S	10	45	52	25	0	MI350 and Xilinx main CoWoS-S customer. AMD also has non-AI projects migrating to 2.5D packaging in 2H26. Likely to be mostly at ASE. Not captured in these forecasts.
CoWoS-L	0	0	0	52	120	CoWoS-L move likely only for MI400 family and beyond, Venice CPUs also to use CoWoS-L.
Broadcom	40	70	96	250	400	TPU v7 (Ironwood) and v8x most of the volume in 2026 and 2027, with small volume for v9 in late 2027. MTIA CoWoS ramp seems slower, OpenAI in 2027. Tomahawk 6 also using CoWoS-S.
CoWoS- S at TSMC	40	70	91	220	220	CoWoS-S is still mainstream in 2026, but more adoption of CoWoS-L from 2026 for Meta, OpenAI and TPUv9 in late 2027
CoWoS -S outside TSMC	0	0	5	10	100	Outsourcing starting to pick up - primarily Amkor for Networking projects based on CoWoS-S. In 2027, some TPU projects may also see some OSAT suppliers emerge - based on technology qualification.
CoWoS-L	0	0	0	20	80	Starting to use CoWoS-L for some projects including advanced TPU models; also likely to incorporate SoIC for some advanced ASICs for TPUv9, MTIA and OpenAI Titan2 in 2027/28.
Alchip + AWS	7	14	5	65	155	Slightly pushout for Trn3 in 2026, but upside in 2027 as lifetime Trn3 volumes are rising due to strong demand. All of Trn3 CoWoS allocation through AWS or Alchip.
Marvell (incl AWS for Trn2)	2	13	75	35	40	Not seeing Trn3 CoWoS or OSAT allocation. MSFT MAIA 3 to ramp in late 2026 with more growth in 2027. Not seeing a big ramp for other CoWoS projects in 2027.
MediaTek	0	0	0	18	60	2026 TPUv8x volume starting, despite multiple small design revisions. Slight upside to 2027 CoWoS forecasts, but depends on execution. TPUv9 Humufish still uncertain on allocation given usage of Intel EMIB and straight-up competition with Broadcom.
Others	5	16	24	45	75	Includes Cisco, Microsoft, GUC, Socionext, other ASIC design houses and smaller startup customers etc.; some non-AI chips also using CoWoS (networking chips for various vendors, some small volume server CPUs etc.).
Overall Total CoWoS Consumption	**134**	**350**	**652**	**1,165**	**1,850**	
yoy		*161%*	*86%*	*79%*	*59%*	
CoWoS Consumption by customer (k wafer per year)	**2023**	**2024**	**2025**	**2026E**	**2027E**	**JPM Comments**
Total TSMC	**134**	**290**	**632**	**1,120**	**1,630**	TSMC includes sub-contracting of oS to ASE/SPIL from 2025
Total Others (OSATs)	0	60	20	45	220	Mainly ASE/SPIL and Amkor, non-TSMC utilization uneven in 2025, given move to CoWoS-L; more full-stack CoWoS from 2H26. Intel EMIB not modeled here, but main volume ramp only in 2028.
Overall Total CoWoS Consumption	**134**	**350**	**652**	**1165**	**1850**	

Source: J.P. Morgan estimates, Company data. Note: Capacity allocation is before adjusting for yield losses, which may vary by accelerator and package type.

Table 2: WMCM forecasts

WMCM Consumption (K wafer per year)	2023	2024	2025	2026E	2027E	JPM Comments
Apple	0	0	0	368	691	All high-end iPhone 18 models in 2H26E using WMCM.
Number of iPhones (M units)				74	138	Likely to adopt for A20 Pro based models on N2. Encompassing iPhone foldable, Pro and Pro max.
Utilization				100%	100%	
Gross Die per wafer				200	200	Expect Package size to be 50% larger than InFO with ~200 dies per Wafer due to side by side stacking of LPDDR5.
% Yield				98%	98%	
TSMC WMCM Capacity (k wafer per year)				**375**	**705**	

Source: J.P. Morgan estimates, Company data.

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Table 3: GPU and AI accelerators chip-level shipment forecasts

GPU and Accelerator Units (M)	2023	2024	2025	2026E	2027E
NVDA					
Hopper Series	1.7	4.3	1.0	0.3	-
Blackwell Series	-	0.2	5.2	5.8	1.0
Rubin Series	-	-	-	2.2	6.1
Rubin Ultra Series (2 die equivalent)	-	-	-	-	2.0
Total NVDA GPU Units	**1.7**	**4.5**	**6.2**	**8.3**	**9.1**
NVDA eq GPU die units	*1.7*	*4.7*	*11.4*	*16.2*	*18.2*
Vera CPU	-	-	-	0.5	2.5
AMD					
MI300/325	0.1	0.5	0.4	-	-
MI350	-	-	0.3	0.2	-
MI450	-	-	-	0.4	0.8
Total AMD GPU Units	**0.1**	**0.5**	**0.7**	**0.6**	**0.8**
Google TPUs					
v4	0.7	0.4	-	-	-
v5	0.5	1.3	-	-	-
v6 /v7 (Trillium/Ironwood)	-	-	1.7	2.7	1.8
v8x - Zebrafish - Mediatek	-	-	-	0.4	1.5
v8ax - Sunfish - Broadcom	-	-	0.1	1.2	3.6
v9 - Pumafish - Broadcom	-	-	-	-	0.1
Total TPU Units	**1.2**	**1.7**	**1.8**	**4.3**	**6.9**
AWS					
Inferentia 2	0.3	0.5	0.1	-	-
Trainium 2	-	0.3	1.5	0.6	-
Trainium 3	-	-	-	1.3	3.1
Total Inferentia / Trainium Units	**0.3**	**0.8**	**1.5**	**1.9**	**3.1**
Meta					
MTIA v2T (Athena)				0.3	0.2
MTIA v3 (Iris)				-	0.4
Total MTIA units (with CoWoS)				**0.3**	**0.5**
Microsoft					
MAIA 200 - GUC				0.1	0.1
MAIA 300 - Marvell				0.1	0.8
Total MAIA units				**0.2**	**0.9**

Source: J.P. Morgan estimates, Company data.

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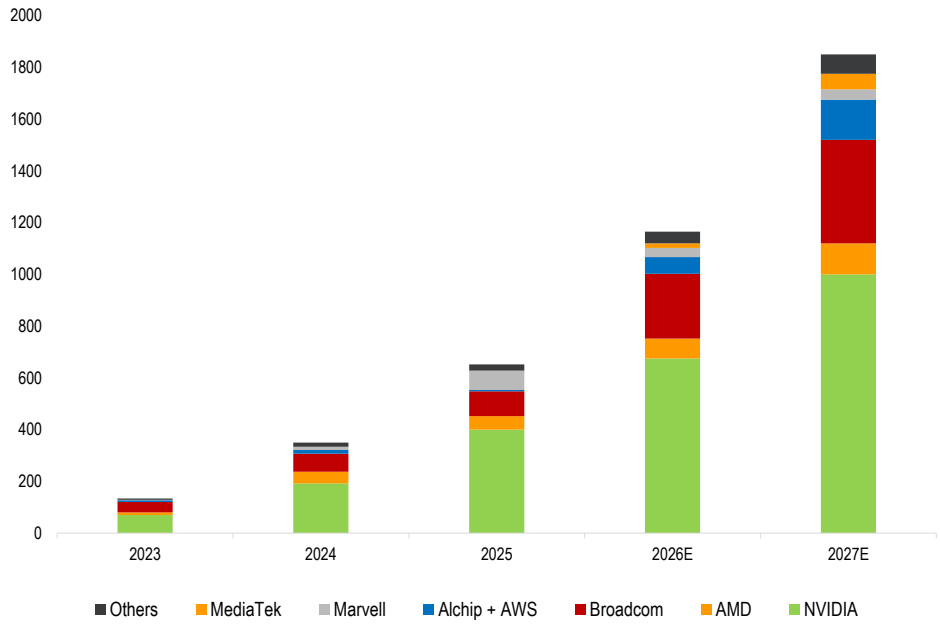
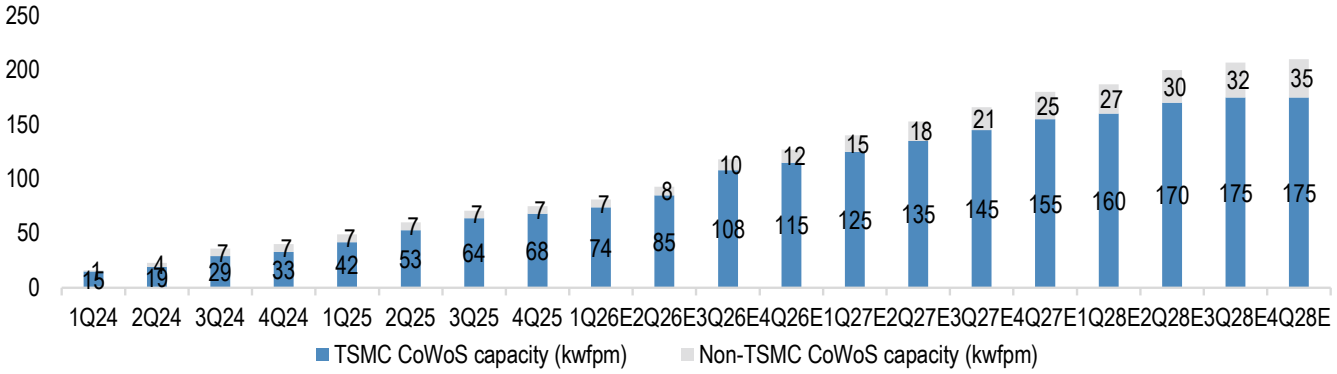
Figure 1: Customer mix for CoWoS

k wafers per year



Source: J.P. Morgan estimates, Company data.

Figure 2: CoWoS capacity – TSMC/non-TSMC



Source: J.P. Morgan estimates, Company data.

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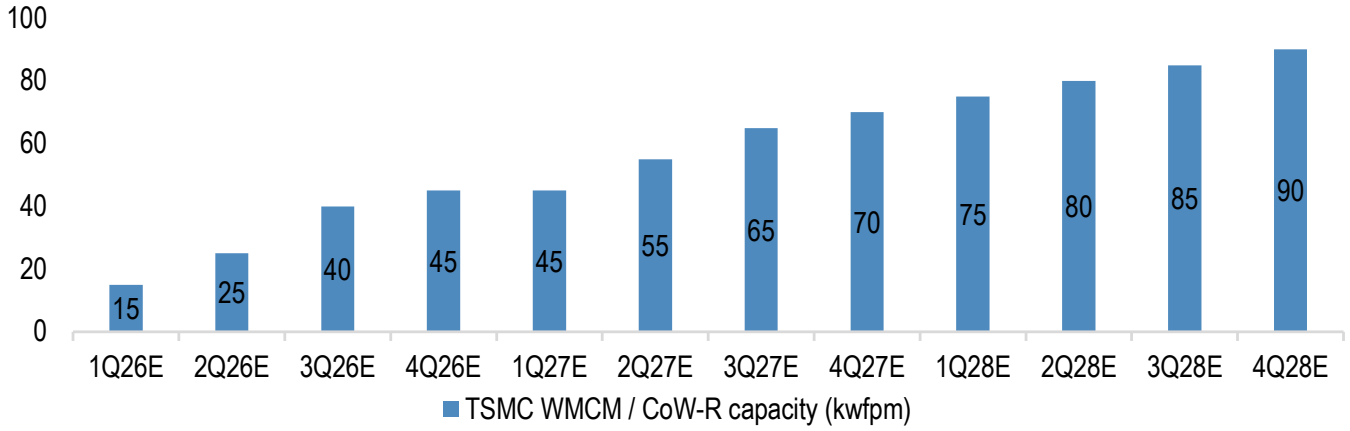
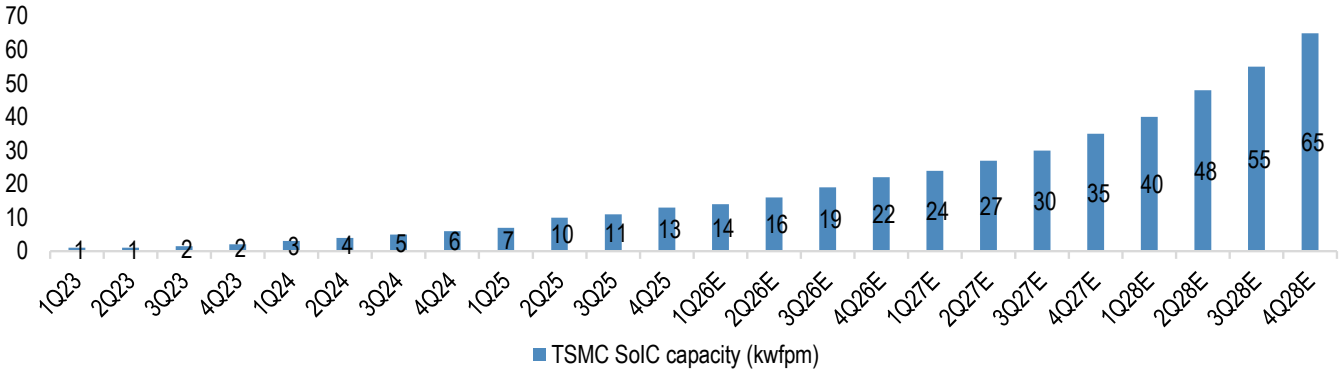
Figure 3: WMCM/CoW-R capacity – TSMC



Source: J.P. Morgan estimates.

Figure 4: SoIC capacity – TSMC



Source: J.P. Morgan estimates, Company data.

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Table 4: Advanced back-end supply chain map for AI ASICs, GPUs and CPUs

Company	GPU/ASIC chip code	MP time	Packaging	Supply chain Partner			Comments
				ASIC	OSAT	Testing	
Nvidia	Blackwell	2H25	CoWoS - L	N.A.	TSMC	KYEC	First product on CoWoS-L
	Rubin	2H26	CoWoS - L	N.A.	TSMC	KYEC, ASE	ASE minority share if qualification for Advantest burn-in oven is completed
	Rubin Ultra	2H27	CoWoS - L	N.A.	TSMC	KYEC, ASE	Likely last NVDA GPU on CoWoS, 2-die-on-interposer default option
	Feynman	2028	Likely CoPoS with SoIC	N.A.	TSMC	KYEC, ASE	Likely first product on CoPoS, should likely use SoIC for high-end versions
	Vera CPU	2H26	CoWoS - R or Equivalent	N.A.	TSMC, ASE, Amkor	KYEC	Start at TSMC, outsourced to ASE and Amkor by 2H26
	GB10 - Digits workstation	3Q25	2.5D Swift	Mediatek IP	Amkor	KYEC	Using Amkor SWIFT 2.5D
	Windows on ARM PC CPU	2026	2.5D packaging	Mediatek IP	Amkor, likely ASE	likely KYEC	ASE likely to break in depending on volumes in 2026
AMD	MI250	2022	EFB	N.A.	Tongfu, ASE	NA	Elevated Fan-out Bridge - Older 2.5D packaging approach
	MI300	2024	CoWoS - S	N.A.	TSMC	handled through TSMC	
	MI350	2025	CoWoS- S	N.A.	TSMC	handled through TSMC	
	MI450	2H26	CoWoS - L	N.A.	TSMC	handled through TSMC	
	Venice CPU	2026	CoWoS - L or Equivalent	N.A.	TSMC start, eventually ASE	ASE	ASE from late 2026, high volume in 2027
	Other GPU and server CPUs	late 2026/27	EFB, PLP	N.A.	ASE , PTI for PLP	ASE	Bulk of full-process packaging growth for ASE from late 2026. PLP from PTI likely for Gaming GPU
Google	TPU v6e (Trillium)	2H24	CoWoS - S	Broadcom	TSMC	KYEC	
	TPU v7 (Ironwood/ Ghostfish)	3Q25	CoWoS - S	Broadcom	TSMC	KYEC	
	TPU v8ax (Sunfish)	2Q26	CoWoS - S	Broadcom	TSMC, may be Amkor	KYEC	N3P, mostly at TSMC, Amkor may get qualified if supply tightness becomes severe
	TPU v8x (Zebrafish)	4Q26	CoWoS - S	Mediatek	TSMC	KYEC	N3P, all at TSMC for now, unlikely to use EMIB
	TPU v9 (Pumafish)	2028	CoWoS - L / likely SoIC	Broadcom	TSMC	KYEC	2nm class TPUs - likely to use CoWoS-L with SoIC
	TPU v9 (Humufish)	2028	EMIB-T / likely SoIC	Mediatek	TSMC (3D) + Intel, Amkor (2.5D)	Likely KYEC	2nm TPU - likely EMIB with 3DSoIC
Broadcom	Tomahawk 5 series	2024	FC BGA	N.A.	ASE, Amkor	NA	
	Tomahawk 6 series	2026	CoWoS-S	N.A.	Amkor, likely ASE	KYEC	Amkor likely to handle majority, given full capacity at TSMC and priority for TPU
Amazon	Trainium 2 (Cayman)	1Q25	CoWoS-R	Marvell	TSMC	Amkor, ASE, PTI	Major share at Amkor and ASE for testing
	Trainium 2.5 (Cayman)	4Q25	CoWoS-R	Marvell	TSMC	Amkor, ASE, PTI	Major share at Amkor and ASE for packaging, depending on eventual volume
	Trainium 3 (Mariana)	2Q26	CoWoS-R	Mostly Alchip	TSMC / maybe ASE	Amkor, ASE, PTI	ASE to handle packaging for Marvell version if it goes HVM, All 3 vendors in for testing
	Trainium 4 (Maverick)	2027/ 2028	likely CoWoS-L / SoIC	Alchip, Marvell	TSMC	NA	Still early in the design cycle, likely SoIC with CoWoS-L
Microsoft	MAIA 100	1Q24	CoWoS-S	GUC	TSMC	GUC	low volume
	MAIA 200	1Q26	CoWoS-S	GUC	TSMC	GUC	low volume
	MAIA 300	2027	CoWoS-S	Marvell	TSMC	Marvell	Likely ramp in higher volume in 2027
META	MTIA 2T (Athena)	3Q25	CoWoS-S	Broadcom	TSMC	NA	Not clear on volume ramp for this product - 5nm based version
	MTIA 3 (Iris)	2H26	CoWoS-L	Broadcom	TSMC	NA	Volume ramp appears slower than anticipated
	MTIA (Arke)	1H27	CoWoS-L with SoIC ?	Broadcom	TSMC	Broadcom (JPMe)	Project seems to have returned to Broadcom after considering Mediatek / Marvell etc
	MTIA 4 (Phoebe)	2028	CoWoS-L + SoIC	Broadcom	TSMC	Broadcom (JPMe)	2nm class project with SoIC adoption , lower grade version compared to original Olympus
OpenAI	Titan / Nexus 2nd Gen	2H27	CoWoS-L + SoIC	Broadcom	TSMC	NA	Not much details on timeline and detailed supply chain
Tesla	Dojo v1	2023	InFO SoW	Alchip	TSMC	NA	First use of InFo SoW
	Dojo v2	TBC	InFO SoW	GUC	TSMC	NA	Project cancelled
	AI5	2027	LPDDR5	GUC	TSMC, maybe Amkor	NA	Likely ramp in 1H2027, supply chain still not fully clear

Source: J.P. Morgan estimates, Company data.

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Table 5: Glossary of important terms

HBM	High Bandwidth Memory	HBM is a high-speed computer memory interface for 3D-stacked synchronous dynamic random-access memory (SDRAM) from Samsung, Micron and SK Hynix.
CoWoS	Chips-on-Wafer-on-Substrate	TSMC's 2.5D interposer-based packaging technology for high-performance computing applications.
CoWoS-S	Chips-on-Wafer-on-Substrate with Silicon Interposer	A CoWoS technique using silicon interposer as the bridge between chip and substrate.
CoWoS-R	Chips-on-Wafer-on-Substrate with Redistribution Layer Interposer	A CoWoS technique using redistribution layer interposer (RDL) to replace silicon interposer, enabling higher packaging size flexibility and lower cost.
CoWoS-L	Chips-on-Wafer-on-Substrate with Local Silicon Interconnect	A CoWoS technique using local silicon interconnect (LSI) to connect dies at high speed rate, while the other area on the chip will use RDL.
COW-R	Chip-on-Wafer with Redistribution Layer Interposer	The first part of the CoWoS process, where the logic chip and HBMs are attached to the RDL interposer.
RDL	Redistribution Layer	A layer of metal interconnects that redistributes I/O access across the chip, facilitating flexible and efficient chip-to-chip connections, particularly in advanced packaging techniques like 2.5D and 3D integration.
WMCM	Wafer-level Multichip Module	An approach that integrates multiple chips (CPU, GPU, memory, etc.) at the wafer stage before slicing them into individual modules. Unlike traditional InFO packaging, focusing on single-die integration, WMCM enables more flexibility in combining different components while maintaining a small footprint.
XPU	X processing unit	The "X" in "XPU" stands for any compute architecture (CPU, GPU, ASIC etc.) that is suitable for different applications.
TPU	Tensor processing unit	Google Cloud TPUs are customized AI accelerators, optimized for training and inference of AI models.
LPDDR	Low Power Double Data Rate	Also known as LPDRAM, LPDDR is used in devices requiring speedy, high-bandwidth data rates without compromising power efficiency.
CPO	Co-Packaged Optics	An integration of optical components and ASICs on a single packaged substrate, enabling high-speed data transmission and less power loss.
ASIC	Application-Specific Integrated Circuit	ASICs (application-specific integrated circuits) are customized chips for certain special purposes, and right-of-use is limited to the chip designer only.
CSP	Cloud Service Provider	A third-party company that provides a cloud-based platform for businesses to access on demand over a network, including compute, storage, and infrastructure services.
FOCoS	Fan Out Chip on Substrate (similar to TSMC's CoWoS)	A fan-out package flip-chip placed on a high pin count ball grid array (BGA) substrate. The fan-out package has an RDL allowing the construction of shorter die-to-die (D2D) interconnections between chips.
Trn	Trainium	Trainium are chips developed by AWS specifically for AI training and inference, which deliver high performance while reducing costs.
InFO	Integrated Fan-Out	A wafer-level integration technology features high-density RDL and Through InFO Via (TIV) for enhanced interconnect performance. It supports various 2D and 3D package schemes optimized for mobile and high-performance computing.
EIC-PIC	Electronic/Photonic Integrated Circuit	EICs utilize electronic components (transistors, capacitors) for electrical signal processing, while PICs use optical components (waveguides, lasers) for light manipulation in data transmission. EICs handle electrons, and PICs handle photons.
CoPoS	Chip-on-Panel-on-Substrate	TSMC's CoPoS is a rectangular substrate evolution of CoWoS, offering more usable space than traditional round wafers, enhancing output efficiency and reducing costs.

Source: J.P. Morgan.

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Investment Thesis, Valuation and Risks

TSMC *(Overweight; Price Target: NT$2400.0)*

Investment Thesis

We expect TSMC's structural growth drivers to remain very strong as leading-edge supply stays tight, with recent AI compute upside further extending N3/N2 constraints into 2027. We believe the company is positioned for strong 1Q26/2Q26 GM beats (~67%), supported by >100% utilization on N3/N5, higher expedited (hot/super-hot) wafer demand with ASP premiums, like-for-like pricing, cross-fab optimization and a weaker TWD. We model FY26/FY27 USD revenue growth of ~35%/~30%, driven by sharp N3 growth and sustained strength in advanced packaging (CoWoS/SoIC). While client PC/Android demand remains weak, we expect upside in HPC and resilient Apple builds to keep leading-edge utilization elevated. We rate the stock OW.

Valuation

Our Dec-26 PT of NT$2,400 is based on a ~20x 12-month forward P/E and reflects the positive fundamental demand driver and our expectation for the recent strong outperformance to continue in the next few quarters. Our target multiple is higher than TSMC's five-year average historical multiple.

Risks to Rating and Price Target

Key downside risks include: (1) the debate on the duration of the AI capex growth cycle; and (2) any negative impact from weak PC/smartphone demand in 2H26.

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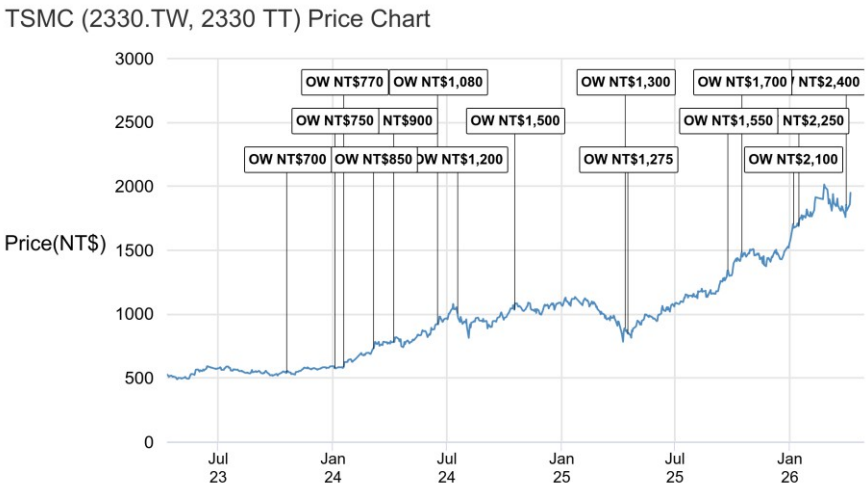

Date	Rating	Price (NT$)	Price Target (NT$)
20-Oct-23	OW	546.0	700
04-Jan-24	OW	578.0	750
19-Jan-24	OW	588.0	770
06-Mar-24	OW	730.0	850
07-Apr-24	OW	780.0	900
17-Jun-24	OW	922.0	1,080
19-Jul-24	OW	1005.0	1,200
18-Oct-24	OW	1035.0	1,500
13-Apr-25	OW	889.0	1,300
18-Apr-25	OW	847.0	1,275
24-Sep-25	OW	1340.0	1,550
17-Oct-25	OW	1485.0	1,700
07-Jan-26	OW	1705.0	2,100
16-Jan-26	OW	1690.0	2,250
02-Apr-26	OW	1855.0	2,400

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
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IB clients**	83%	79%	74%
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